1. If M represents the median and D the mode of the measurements 5, 9, 3, 5, 8 then (M,D) is

A. (6,5) B. (5,8) C. (5,7)
D. (5,5) E. (7,5)

2. A construction company is owned by two partners X and Y and it is agreed that their profit will be divided in the ratio 4:5. at the end of the year. Y received #5,000 more than x. what is the total profit of the company for the year?

A. #20,000.00 B. P'0#25,000.00 C. #30,000.00
D. #15,000.003 E.#45,000.00

3. Given a regular hexagon, calculate each interior angle of the hexagon.

A. 60^0 B. 30^0 C. 120^0
D. 45^0 E. 135^0

4. Solve the following equations
$4x - 3 = 3x + y = 2y + 5x - 12$

A. $4x=5, y=2$ B. $x=2, y=5$ C. $x=-2, y=-5$
D. $x=5, y=-2$ E. $x=-5, y=-2$

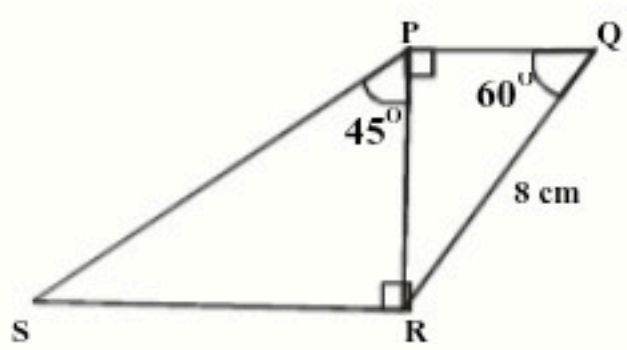
5. If $x = 1$ is root of the equation
$x^3 - 2x^2 - 5x + 6$, find the other roots

A. -3 and 2 B. –2 and 2 C. 3 and –2
D. 1 and 3 E. –3 and 1

6. If x is jointly proportional to the cube of y and the fourth power of z. In what ratio is x increased or decreased when y is halved and z is doubled?

A. 4:1 increase B. 2:1 increase C. 1:4 decrease
D. 1: 1 no change E. 3: 4 decrease

7.



In the above figure PQR = 60^0, QPR = 90^0, PRS = 90^0, RPS = 45^0, QR= 8cm. Determine PS

A. $2\sqrt{3}$cm B. $4\sqrt{6}$cm C. $2\sqrt{6}$cm
D. $8\sqrt{6}$cm E. 8cm

8. Given that cos z = L, where z is an acute angle find an expression for $\frac{Co + Z - cosecz}{sec Z + tan z}$

A. $\frac{1 - L}{1+L}$ B. $\frac{L^2 - \sqrt{1-L^2}}{L2+L-1}$ C. $\frac{-L-\sqrt{1-L}}{(C1+L) + \sqrt{1-L^2}}$

D. $\frac{\sqrt{L-1}}{(L1+L^2) + \sqrt{1-L^2}}$ E. $\frac{L-(L^2-1)}{1+\sqrt{1-L^2}+\sqrt{1-L^2}}$

9. If $0.0000152 \times 0.00042 = A \times 10^8$, where $1 £ A < 10$, find A and B.

A. A = 9, B = 6`.38 B. A = 6.38, B = -9 C. A = 6.38, B = 9
D. A = 6.38, B = -1 E. A = 6.38, B = 1

10. If x + 2 and x – 1 are factors of the expressions $lx + 2kx^2 + 24$, find the values of l and k

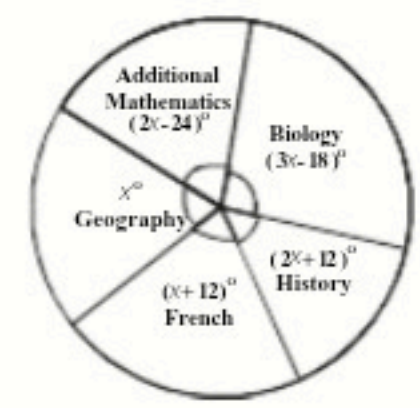
A. l=-6, k=-9 B. l=-2, k=1 C. l=-2, k=-1
D. l=0, k=1 E. l=6, k=0

11. Make T the subject of the equation

$$\frac{av}{1-V} = \sqrt[3]{\frac{2V + T}{a \quad 2T}}$$

A. 3av/(1-v) B. $2v(1-v)^2 - a^2v^2/2a^2v^2 - (1-V)^2$
C. $2v(1 - v)^2 + a^3v^2/ 2a^2v^2 + (1 - v)^2$
D. $2v(1 - v)^2 - a^4v^3/2a^3v^3 - (1 - v)^3$
E. $2v(1-v)^3 - a^4v^3/2a^3v^3 + (1-v)^3$

12.



In a class of 60 pupils, the statistical distribution of the number of pupils offering Biology, History, French, Geography and Additional Mathematics is as shown in the pie chart above. How many pupils offer Additional Mathematics?

A. 15 B. 10 C. 18
D. 12 E. 28

13 The value of $(0.303)^3 - (0.02)^3$ is

A. 0.019 B. 0.0019 C. 0.00019
D. 0.000019 E. 0.000035

14. y varies partly as the square of x and y partly as the inverse of the square root of x. write down the expression for y if y = 2 when x = 1 and y = 6 when x = 4

A. $y = \frac{10x^2}{31} + \frac{52}{31\sqrt{x}}$ B. $y = x^2 + \frac{1}{\sqrt{x}}$
C. $y = x^2 + \frac{1}{x}$ D. $y = \frac{x^2}{31} + \frac{1}{31\sqrt{x}}$ E. $y = \frac{10}{31}(x^2 + \frac{1}{\sqrt{x}})$

15. Simplify $(x - 7) / (x^2 - 9) (x^2 - 3x)/(x^2 - 49)$

A. x/(x-3)(x+7) B. (x+3)(x+7)/x C. x/(x-3)(x-7)
D. x/(x+3)(x+7) E. x/(x+4)(x+7)

16. The lengths of the sides of a right-angled triangle at (3x + 1)cm, (3x - 1)cm and x cm.

A. 2 B. 6 C. 18
D. 12 E. 0

17. The scores of a set of a final year students in the first semester examination in a paper are 41,29,55,21,47,70,70,40,43,56,73,23,50,50. find the median of the scores.

A. 47 B. $48^1/_2$ C. 50
D. 48 E. 49

1. If M represents the median and D the mode of the measurements 5, 9, 3, 5, 8 — **A:** (6,5); **B:** (3,5); **C:** (5,7); **D:** (5,5); **E:** (7,5)

2. A construction company is owned by two partners X and Y and it is agreed that their profit will be divided in the ratio 4:5. At the end of the year, Y received ₦5,000 more than X. What is the total profit of the company for the year? — **A:** ₦20,000.00; **B:** ₦25,000.00; **C:** ₦30,000.00; **D:** ₦15,000.00; **E:** ₦45,000.00

3. Given a regular hexagon, calculate each interior angle of the hexagon — **A:** 60°; **B:** 30°; **C:** 120°; **D:** 45°; **E:** 135°

4. Solve the following equations: $4x - 3 = 3x + y - 2y + 5x - 12$ — **A:** x = -5, y = 2; **B:** x = 2, y = 5; **C:** x = -2, y = -5; **D:** x = 5, y = -2; **E:** x = -5, y = -2

5. If x = 1 is a root of the equation $x^2 - 2x - 3x + 6$, find the other roots — **A:** 3 and 2; **B:** -2 and 2; **C:** 3 and -2; **D:** 1 and 3; **E:** -3 and 1

6. If x is jointly proportional to the cube of y and the fourth power of z. In what ratio is x increased or decreased when y is halved and z is doubled? — **A:** 4:1 increase; **B:** 2:1 increase; **C:** 1:4 decrease; **D:** 1:10 change; **E:** 3:4 decrease

7. (Triangle diagram) In the above figure $\angle PQR = 60^\circ$, $\angle QPR = 90^\circ$, $\angle PRS = 90^\circ$, $\angle RPS = 45^\circ$, QR = 8 cm. Determine PS — **A:** $2\sqrt{3}$ cm; **B:** $4\sqrt{2}$ cm; **C:** $2\sqrt{6}$ cm; **D:** $8\sqrt{2}$ cm; **E:** 8 cm

8. Given that $\cos z = L$, where z is an acute angle find an expression for $\cos z - \csc z$ — **A:** $\sec z - \tan z$; **B:** $L^2 - \sqrt{1 - L^2}$; **C:** $\frac{L - \sqrt{1 - L^2}}{1 + L}$; **D:** $\frac{L - \sqrt{1 - L^2}}{L(1 + L)}$; **E:** $\frac{L - \sqrt{1 - L^2}}{1 + \sqrt{1 - L^2}}$

9. If $0.0000152 \times 0.00042 = A \times 10^f$, where $1 \le A < 10$, find f and A — **A:** $A = 9.8, f = -8$; **B:** $A = 6.38, f = -9$; **C:** $A = 6.38, B = 9$; **D:** $A = 6.38, B = -1$; **E:** $A = 6.38, B = -8$

10. If $x + 2$ and $x - 1$ are factors of the expression $[x + 2x^2 + 24]$, find the values of k and l — **A:** $l = 0, k = 9$; **B:** $l = -2, k = 1$; **C:** $l = 2, k = 9$; **D:** $l = 0, k = 1$; **E:** $l = 0, k = 0$

11. Make T the subject of the equation $a - v = \frac{\sqrt{2V/T}}{2T}$ — **A:** $3a\sqrt{(1 - v)}$; **B:** $2v(1 - v^2) - a^2\sqrt{2v^2} - (1 - V)^2$; **C:** $2v(1 - v^2) - a^2\sqrt{2v^2} - (1 - v)^2$; **D:** $2v(1 - v^2) - a^2\sqrt{2v^2} - (1 - v)^2$; **E:** $2v(1 - v^2) - a^2\sqrt{2v^2} - (1 - v)^2$

12. (Pie chart) In a class of 60 pupils, distribution of pupils offering Biology, History, French, Geography and Additional Mathematics is shown. How many pupils offer Additional Mathematics? — **A:** 6; **B:** 10; **C:** 18

13. The value of $(0.303)^2 - (0.02)^2$ is — **A:** 0.019; **B:** 0.0019; **C:** 0.00019

14. y varies partly as the square of x and partly as the square root of x. If $y = 1$ when $x = 1$ and $y = 6$ when $x = 4$, find the values of constants a and b. — **A:** $y = ax^2 + b\sqrt{x}$; **B:** $y = ax + b\sqrt{x}$; **C:** $y = x(ax + b)$; **D:** $y = x(a + b\sqrt{x})$; **E:** $y = x^2(a + b\sqrt{x})$

15. Simplify $\frac{x - 7}{x^2 - 9} \div \frac{x^2 - 3x}{x^2 - 49}$ — **A:** $x(x - 3)(x + 7)$; **B:** $(x + 3)(x - 7)$; **C:** $x(x - 3)(x + 7)$; **D:** $x(x + 3)(x + 7)$; **E:** $x(x - 3)(x + 7)$

16. The lengths of the sides of a right-angled triangle are $(3x + 1)$ cm, $(3x - 1)$ cm and x cm. (Find x) — **A:** 2; **B:** 6; **C:** 8; **D:** 12; **E:** 10

17. The scores of final year students are 41, 29, 55, 21, 47, 70, 40, 46, 38, 73, 23, 50. Find the median — **A:** 48; **B:** 47; **C:** 49; **D:** 50; **E:** 45

If you want the correct answers computed and listed beside each question, I can solve them and return the results in the same one-line format. Which would you prefer: **(A)** extract only (as above), or **(B)** extract plus compute correct answers?

1. D — Median = 5, Mode = 5 so (M,D)=(5,5)\. **2.** **E** — Ratio \(4:5, difference $= k$ gives $k = 5000$; total $= 9k = 45{,}000$.

3. C — Interior angle $= \dfrac{(6-2)\cdot 180}{6} = 120^\circ$.

4. No option matches — Equation reduces to $y = 4x - 9$; infinitely many solutions, not a single pair.

5. A — Polynomial $x^2 - 5x + 6 = (x-2)(x-3)$; other roots 2 and 3.

6. B — Factor $= \left(\frac{1}{2}\right)^3 \cdot 2^4 = 2$; x doubles (increase 2: 1).

7. No option matches — Diagram details are insufficient to determine PS uniquely from the text.

8. No option matches — $\cos z - \csc z = L - \dfrac{1}{\sqrt{1-L^2}}$, which does not equal any listed choice.

9. B (closest) — Product $= 6.384 \times 10^{-9}$; $A \approx 6.384$, $f = -9$ (closest listed: 6.38, -9).

10. No option matches — Problem statement/transcription is ambiguous; cannot determine k, l reliably.

11. No option matches (formula given) — Rearranged: $T = \left(\dfrac{\sqrt{2V}}{2(a-v)}\right)^{2/3}$.

12. No option matches — Pie-chart sector sizes not provided in text; cannot compute count for Additional Mathematics.

13. No option matches — $(0.303)^2 - (0.02)^2 = 0.091409$; none of the choices equal this.

14. A — $y = ax^2 + b\sqrt{x}$. Solving $a + b = 1$, $16a + 2b = 6$ gives $a = \frac{2}{7}$, $b = \frac{5}{7}$.

15. No option matches — Simplifies to $\dfrac{(x-7)^2(x+7)}{x(x-3)^2(x+3)}$; none of the given options match.

16. D — Taking $3x + 1$ as hypotenuse yields $x(x-12) = 0$; positive solution $x = 12$.

17. No option matches — Sorted data median $= \dfrac{41+46}{2} = 43.5$; not among the choices.

If you want, I can now: **(1)** mark only the questions where a listed option is correct and give a one-line algebraic check, or **(2)** rework any specific question step-by-step so it's clearer. Which would you like?